

RetainIQ — Who to call, and what it costs

- Hansraj Tiwari & swayangjeet nayak
- ChurnZero 26 · IIT Kharagpur

The 80:1 problem

- Missed churner (FN): ₹40,000
- Unnecessary call (FP): ₹500
- Cost ratio 80:1 — threshold 0.5 is wrong for this problem

Dataset snapshot

- 8,101 train · 97 features · 16.1% churn
- 2,026 test customers · submission 16.6% positive
- Inactivity and balance stress separate churners

Where churn concentrates

- Higher churn: low tenure, low digital engagement, complaints
- Lower churn: stable tenure + active digital footprint
- Segments guide playbook priority

RetainIQ pipeline

- LGB + CatBoost OOF → meta on OOF → Platt calibration
- Rank → churn_probability; calibrated + cost → churn_prediction
- IPTW uplift · fairness (gender, region)

Model journey (PR-AUC saturated)

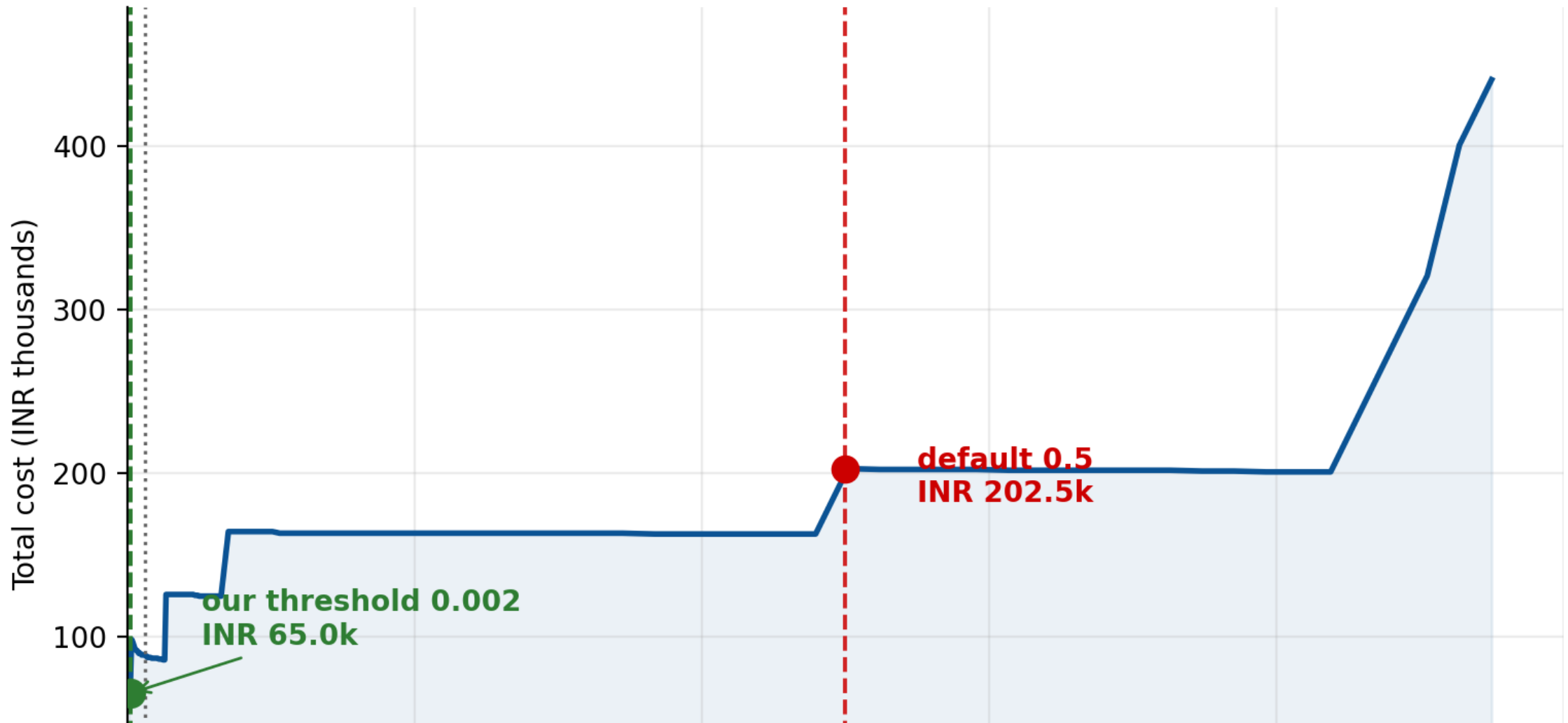
- Logistic → LGB → stack → calibrated: PR-AUC ≈ 0.9999
- AUC gains are flat on this dataset
- Where we win: cost-optimal cutoff + uplift + playbook

Out-of-fold results

	t = 0.002	t = 0.5
PR-AUC	0.9999	0.9999
Business cost (INR)	65,000	202,500
Recall	99.9%	99.6%

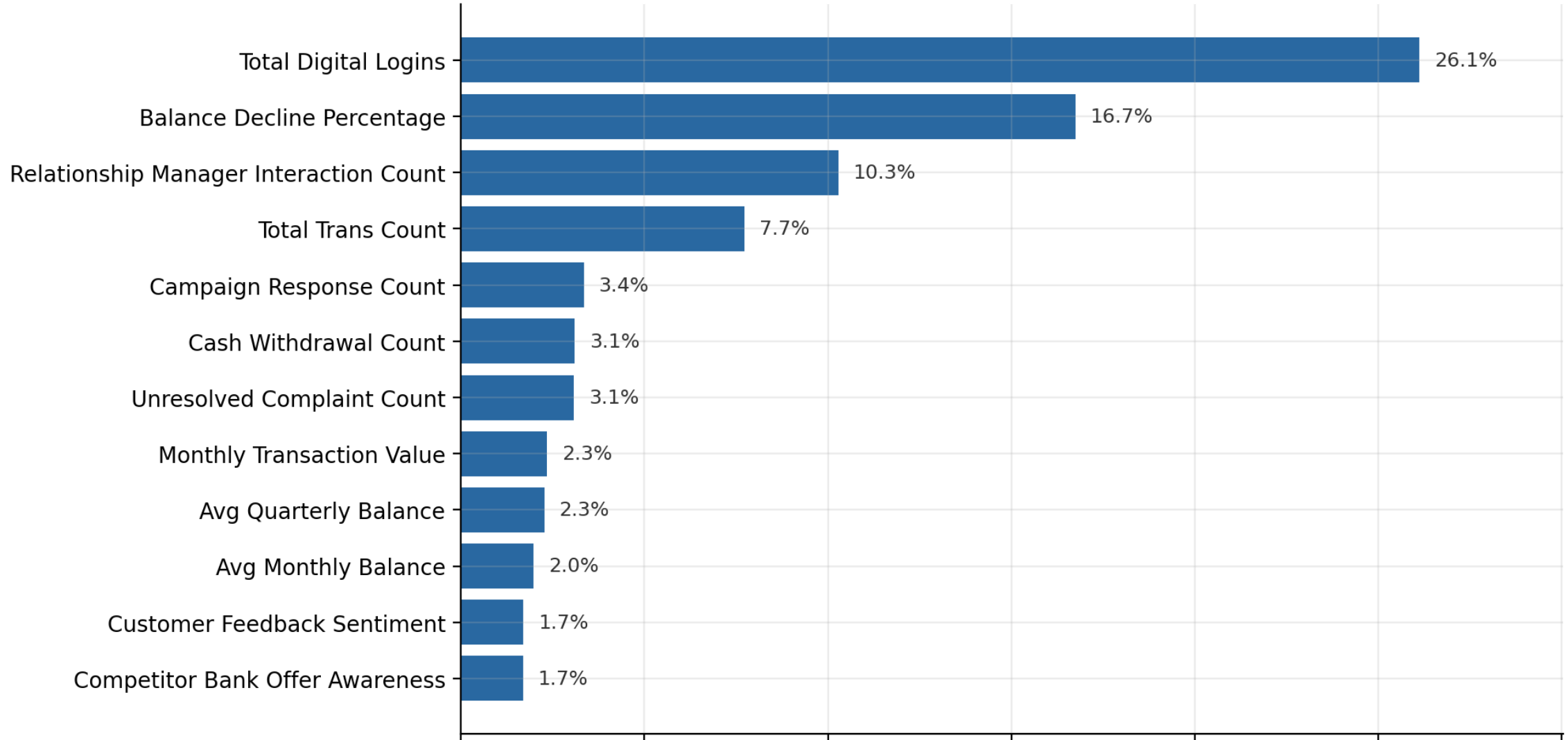
Same model, different cutoff

Same model, different cutoff — INR 138k saved (67.9%)



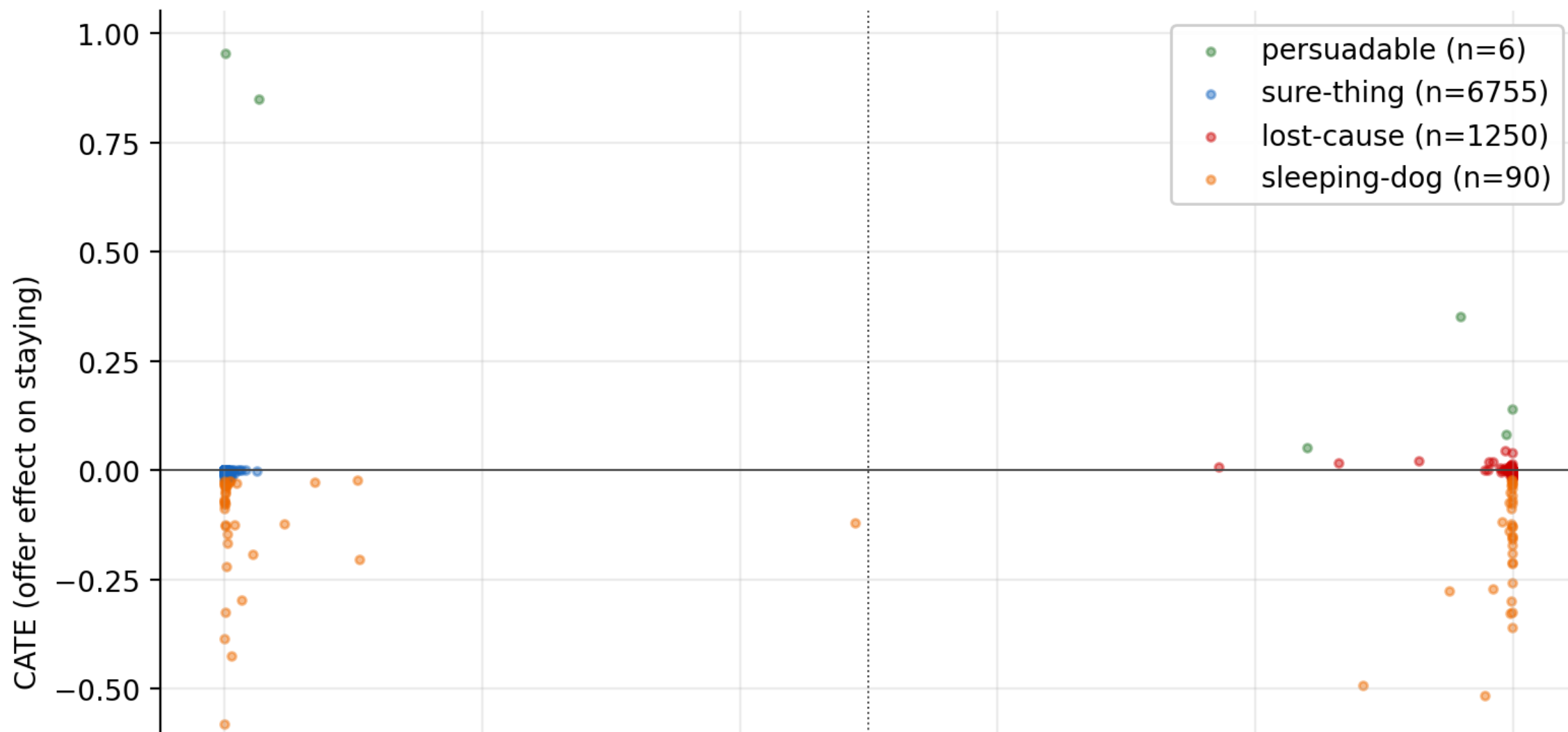
What drives churn

Top features (mean LightGBM gain)



Who is worth calling?

Who is actually worth calling?

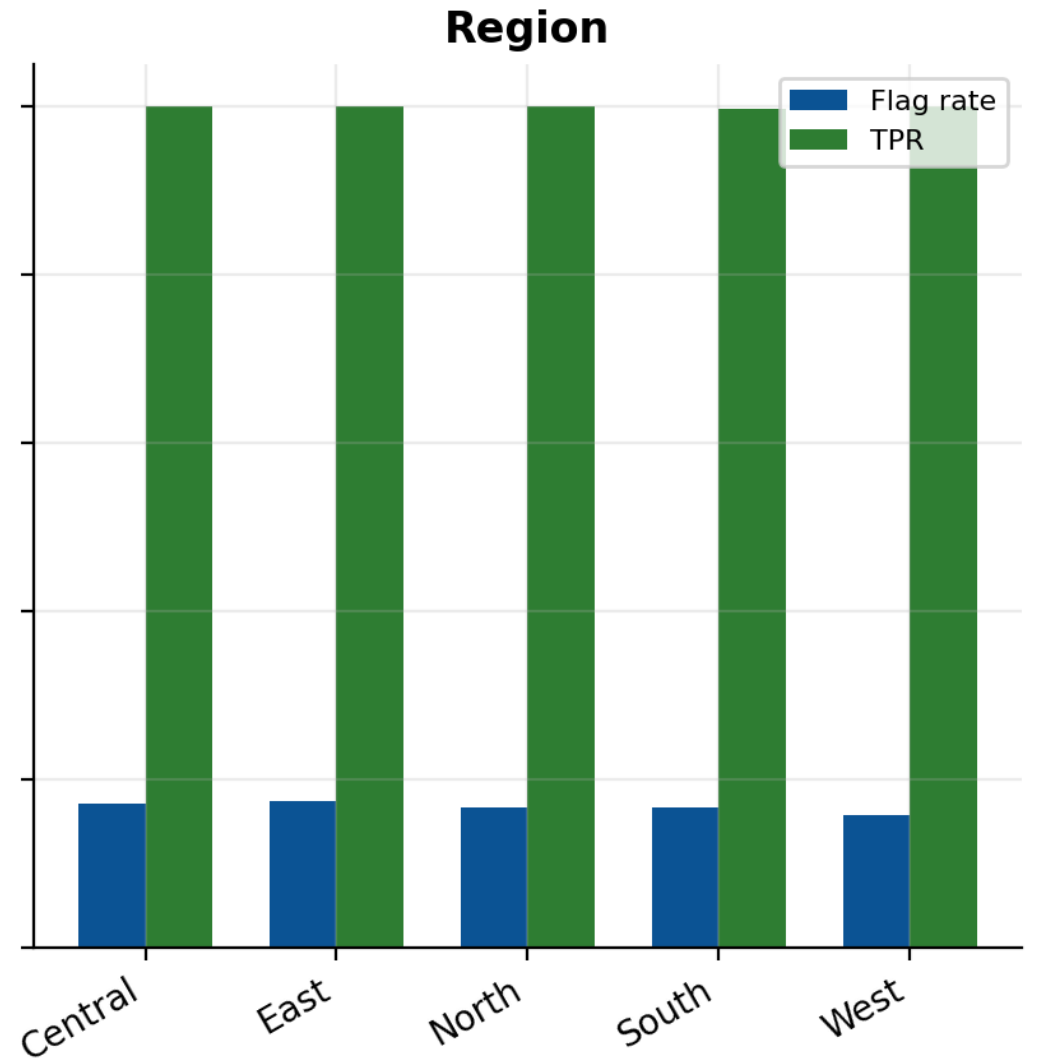
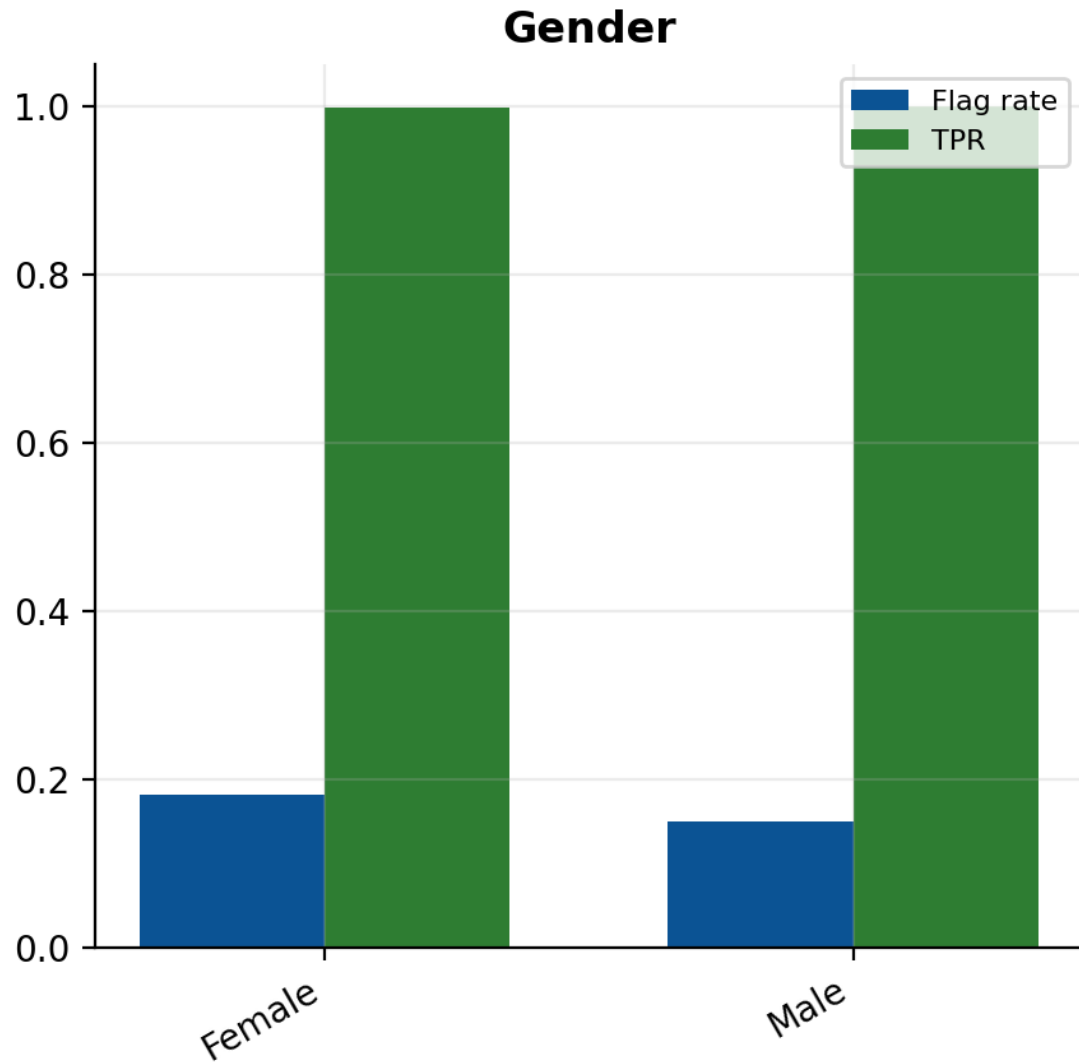


Case study — persuadable segment

- Customer 140348 · segment: persuadable
 - $P(\text{churn}) = 0.959 \cdot \text{CATE} = +0.351$
 - High uplift in persuadable band (CATE + risk), not mass blast
 - Uplift CATE = +0.351: offer strongly associated with staying for this profile.
 - Actions: targeted RM call, resolve complaints, timed waiver—not mass campaigns.

Fairness at operating threshold

Fairness by group



Monday-morning playbook

- Persuadable → RM call + waiver
- Sleeping-dog → no promotional contact
- Lost-cause / sure-thing → low-touch or cross-sell
- ROI: ₹137.5k saved (~₹17/customer)

Production path

- Weekly CRM scores · quarterly retrain · PSI on drivers
- `python -m scripts.train` · 58 pytest tests + CI
- Limits: observational uplift, saturated AUC

RetainIQ — Thank you

- Questions?
- github.com/Stormynubee/retainiq-churnzero-26